

Heinrich Weber
Lehrbuch der Algebra
Working English Translation Batch 12
Alternating, Orthogonal, and Fractional Linear Groups

Translated into modern mathematical English

Translator's note

This batch continues Weber's linear-group chapter and begins the next section on fractional linear substitutions. The available cleaned source skips from §49 to §51; this batch translates all sections present in that interval.

§47. The alternating group and its invariants

The alternating group on n letters consists of all even permutations and has index 2 in the symmetric group. The function

$$\Delta = \prod_{i < k} (x_i - x_k) \tag{1}$$

is an invariant of the alternating group. More precisely, it is an invariant which changes sign under every odd permutation.

The square

$$\Delta^2$$

is therefore an absolute invariant of the symmetric group. It is a symmetric function of the x_i , and is represented by the discriminant of the corresponding equation.

The most general invariant of the alternating group can be written in the form

$$F = F_1 + \Delta F_2, \tag{2}$$

where F_1 and F_2 are symmetric functions, that is, invariants of the full symmetric group.

For $n \geq 5$, the alternating group is simple: it has no normal subgroup other than the identity and itself. This fact is one of the central structural reasons why the general equation of degree at least five cannot be solved by radicals.

§48. Orthogonal substitutions

A linear substitution is called *orthogonal* if it transforms the quadratic form

$$x_1^2 + x_2^2 + \cdots + x_n^2 \tag{3}$$

into itself. For a substitution A with coefficients $a_i^{(k)}$, this means that the identity

$$\sum_k \left(\sum_i a_i^{(k)} x_i \right)^2 = \sum_i x_i^2 \quad (4)$$

holds in the variables x .

Comparing coefficients gives the relations

$$\sum_k a_i^{(k)} a_j^{(k)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (5)$$

For example, if

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix},$$

then the orthogonality relations are

$$a_1^2 + b_1^2 + c_1^2 = 1, \quad a_1 a_2 + b_1 b_2 + c_1 c_2 = 0, \quad (6)$$

$$a_2^2 + b_2^2 + c_2^2 = 1, \quad a_1 a_3 + b_1 b_3 + c_1 c_3 = 0, \quad (7)$$

$$a_3^2 + b_3^2 + c_3^2 = 1, \quad a_2 a_3 + b_2 b_3 + c_2 c_3 = 0. \quad (8)$$

These equations say exactly that the row vectors of A form an orthonormal system.

From these relations one obtains

$$|A|^2 = 1,$$

and hence

$$|A| = \pm 1.$$

The orthogonal substitutions with determinant $+1$ are those of the *first kind*, or rotations. Those with determinant -1 are those of the *second kind*, or reflections.

The totality of all orthogonal substitutions in n variables forms a group, the *orthogonal group*. One may regard either the positions of the body or the rotations themselves as the elements of the group.

§49. Linear fractional substitutions

We shall now show that the group of ternary orthogonal substitutions is isomorphic to the group of linear fractional substitutions in one variable, or equivalently to the group of binary substitutions acting on ratios.

Write a ternary orthogonal substitution as

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3), \quad (9)$$

so that

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2. \quad (10)$$

We first apply a fixed, nonorthogonal change of variables of the form

$$x_1 = \xi_1 + \xi_2, \quad y_1 = \eta_1 + \eta_2, \quad (11)$$

$$x_2 = \xi_1 - \xi_2, \quad y_2 = \eta_1 - \eta_2, \quad (12)$$

$$x_3 = \xi_3, \quad y_3 = \eta_3. \quad (13)$$

Under this transformation the quadratic condition is converted into an equivalent condition in the new variables.

Now introduce binary variables by setting, in effect,

$$x_1 = \sqrt{2} \xi_1 \xi_3, \quad x_2 = \sqrt{2} \xi_2 \xi_3.$$

Then the corresponding quadratic relation vanishes identically. The transformed variables y_1, y_2, y_3 become binary quadratic forms in ξ_1, ξ_2 , and these forms must satisfy the transformed quadratic equation identically.

The product relation then forces the relevant quadratic forms to decompose into linear factors. Since y_2 and y_3 cannot have a common factor without making the determinant of the transformation vanish, the factors are arranged in such a way that the ternary orthogonal substitution is represented by a binary linear substitution, considered projectively.

Thus ternary rotations are represented by fractional linear transformations. In modern language, this is the classical passage from the binary linear group, modulo scalar multiplication, to the rotation group in three dimensions.

§51. Finite groups of linear fractional substitutions; poles of groups

From the preceding developments it follows that, once all finite groups of linear fractional substitutions are known, one immediately obtains all finite groups of proper ternary orthogonal substitutions and all finite groups of binary linear substitutions with determinant 1.

Let S be a finite group of linear fractional substitutions in a variable x . Write its substitutions in canonical form

$$x' = \frac{\alpha x + \beta}{\gamma x + \delta}, \quad (14)$$

with determinant condition

$$\alpha\delta - \beta\gamma = 1.$$

A nonidentity substitution of this form generally has two fixed points, or *poles*, obtained from the quadratic equation

$$\gamma x^2 + (\delta - \alpha)x - \beta = 0. \quad (15)$$

Only in the exceptional case $\gamma = 0$ and $\alpha = \delta$, that is, for a translation $x' = x + \beta$, does the equation reduce to degree one; then there is only one improper pole, situated at infinity.

The pole structure is the key to the classification of finite fractional linear groups. A finite group acts on the sphere of projective values, and the exceptional points with nontrivial stabilizer determine the group.

§52. The cyclic and dihedral groups

The simplest finite groups are the *cyclic groups*. A cyclic group of degree n is generated by the substitution

$$x' = \varepsilon x, \quad (16)$$

where ε is a primitive n -th root of unity. The n substitutions of the group are

$$x, \varepsilon x, \varepsilon^2 x, \dots, \varepsilon^{n-1} x. \quad (17)$$

The two poles are 0 and ∞ , and both are fixed by every substitution of the group.

A *dihedral group* of degree $2n$ contains, besides these substitutions, the n substitutions

$$x' = \frac{\varepsilon^k}{x}, \quad k = 0, 1, \dots, n-1. \quad (18)$$

These interchange the two poles 0 and ∞ . Thus the group consists of the substitutions

$$x' = \varepsilon^k x, \quad x' = \frac{\varepsilon^k}{x}, \quad k = 0, 1, \dots, n-1. \quad (19)$$

The element ε must run through a full system of n -th roots of unity. Hence, if the ε occurring in the generator is primitive, the whole cyclic subgroup is obtained from its powers, and the dihedral group is obtained by adjoining the inversion $x \mapsto 1/x$, together with its products with the rotations.